
DIY Low-Cost Mini Met Station

Shelby Kaplan, Rut Parnas, Ramson Kabenla, & Mary Mitchell Adhiambo

Final presentation for Envirotech: DIY sensors for environmental research

Prof. Elad Levintal

Ben-Gurion University of the Negev · 19 July 2026

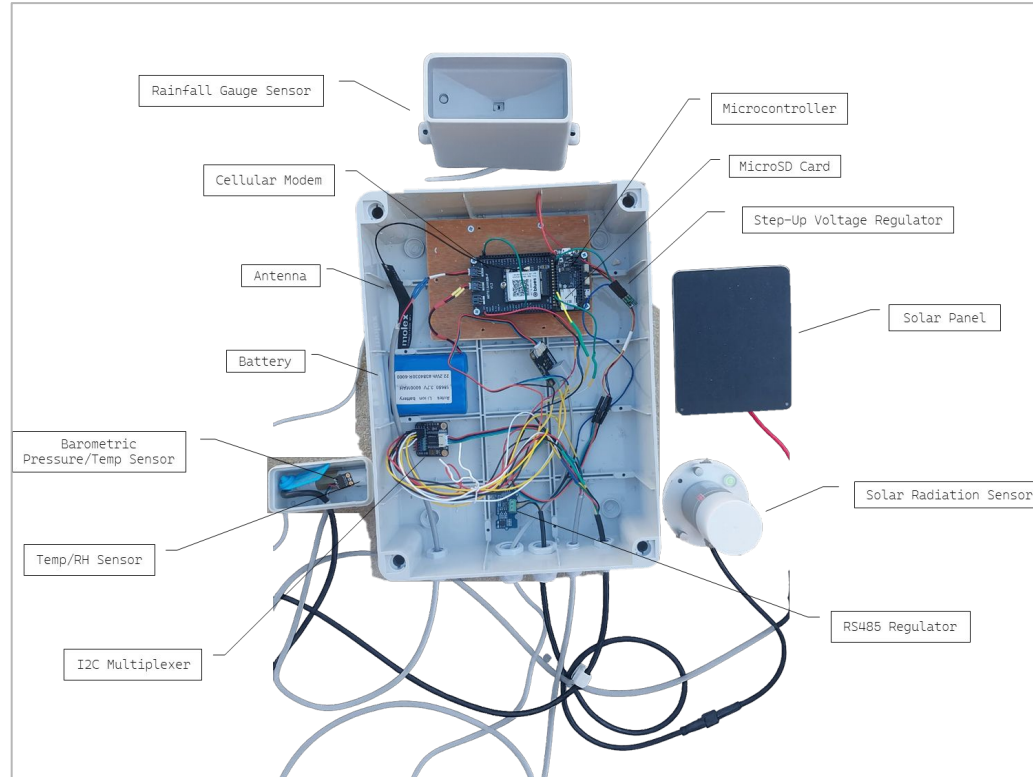
The Problem

- Reliable weather data comes from few, costly fixed stations — coarse spatial coverage, poor local resolution
- Satellite rainfall alternatives lag actual conditions by 1 day – 7 weeks (Maidment et al., 2017)
- Commercial stations cost up to €7,350/unit (Botero-Valencia et al., 2022)
- Result: Smallholder farmers & environmental managers lack localized, real-time data for land/water decisions, networks can't scale

Previous Work

System	Advantages	Disadvantages	Link
Development of radiation weather stations [Lynch et al., 2018]	Used a highly advanced Campbell Scientific CR100 data logger.	Highly expensive Required software customization for each unique device is time-consuming.	https://doi.org/10.1097/hp.0000000000000962
Arduino-based automatic wireless weather station with remote graphical application and alerts [Saini et al., 2016]	Integrated a Labview-based graphical user interface for data display, logging and alert generation	Wireless data transmission was limited to only 1 kilometre.	https://doi.org/10.1109/spin.2016.7566768

Our Solution



Low-cost sensors

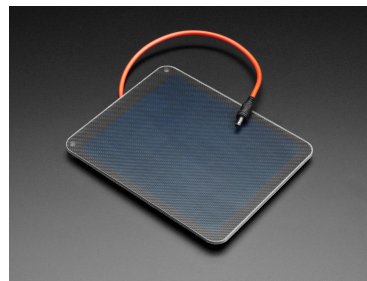
- Temperature (2x)
- Relative humidity
- Rainfall
- Solar radiation
- Pressure

Long-term power

- Solar panel
- Li-ion batteries
- 10-min sleep cycles

Real-time data

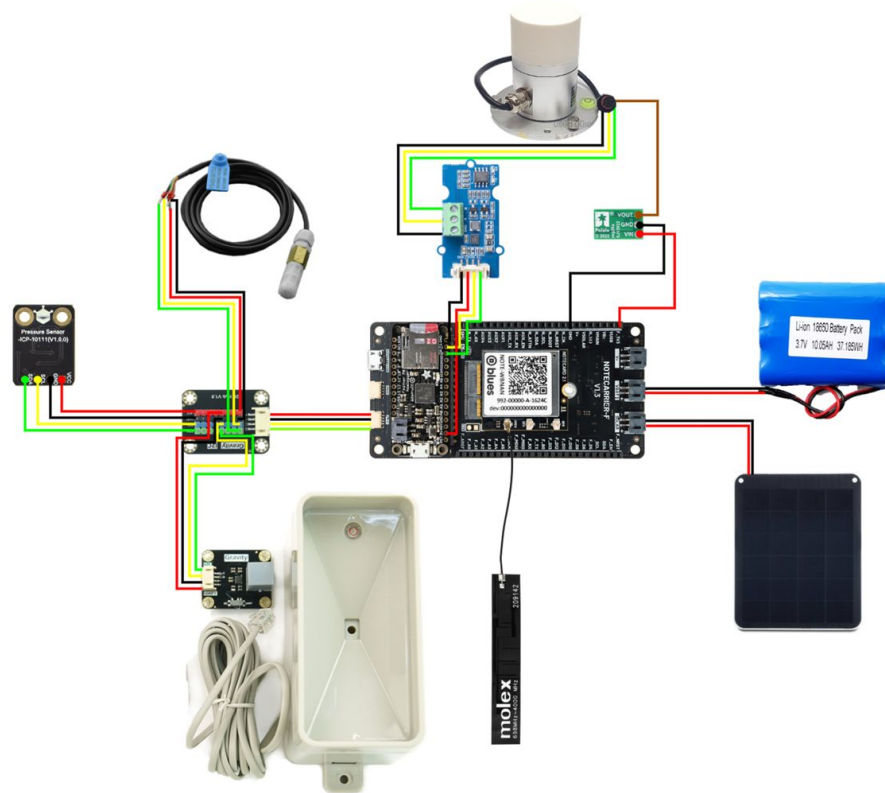
- Cellular modem (Blues Notecard)
- Live cloud dashboard (Notehub)
- Local SD-card backup
- IMS calibration



Bill of Materials

[illegible]

Connection Diagram

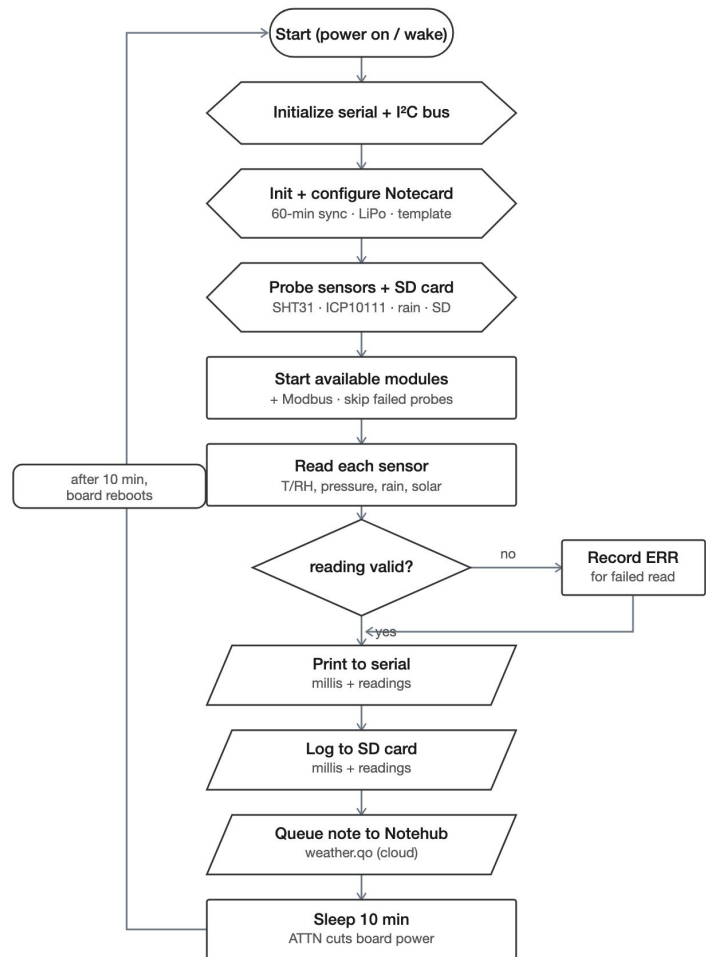


Installation



Software

- One-pass setup() each cycle — no continuous loop
- Reads sensors, logs to SD, queues to cloud, then sleeps
- Wakes every 10 min; cloud syncs every 60 min; failed sensors flagged ERR



Data Pipeline: Device > Cloud > Dashboard

- Notecard queues
10-minute readings locally,
syncs to Notehub hourly
- Custom Python script pulls
via Notehub + IMS APIs
- IMS Sde Boker/Ashalim
stations added as
ground-truth reference
- Output feeds directly into
"nearly-live" dashboard

blues notehub

Stream

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Setup Plugins

Enviroment Mini Met Station

03/10/2026 - 03/10/2026

03/10/26

Filter events

Showing 10/10 selected

All Events

View

Route

Delete

Transform

	Status	Transport	Component	Host Location	Host URI	File	Body
▼	▼	▼	▼	▼	▼	▼	▼
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Device

Stream

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Stream

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Setup Plugins

Download Result as: [CSV](#) [JSON](#) [XLSX](#)

Station	Date - Time	Relative Humidity (%)	Temp., (C)	Humid., (mm)	Wind direction	Wind speed (km/h)	Wind speed (m/s)	1 minute wind speed (km/h)	10 minute wind speed (km/h)	Time ending (mm)	Wind direction	Stand. deviat., wind direction	Rainfall (mm)
Sea Breeze	14/07/2018	85	22.5	22.4	22.2	306	376	2.7	3.4	3:57	23	46.3	0
Sea Breeze	14/07/2018	85	22.2	22.5	22.2	303	294	2.7	3.3	2:8	00:03	4	17.4
Sea Breeze	14/07/2018	85	22.1	22.4	22	300	320	2.2	3.1	2:7	00:19	1	8.2
Sea Breeze	14/07/2018	86	22.1	22	21.9	305	307	2.1	2.9	0:22	00:25	3.4	10.4
Sea Breeze	14/07/2018	87	21.9	21.9	21.8	312	356	1.9	2.6	2:3	00:31	3.3	10.1
Sea Breeze	14/07/2018	88	21.8	21.9	21.6	319	302	1.7	2.4	1:9	00:42	2.8	10.9
Sea Breeze	14/07/2018	89	21.6	21.8	21.6	314	318	1.9	2.6	1:7	00:49	2.9	10.4
Sea Breeze	14/07/2018	89	21.6	21.6	21.6	321	396	1.8	1.9	1:7	01:06	2.3	8.8
Sea Breeze	14/07/2018	90	21.5	21.6	21.4	313	304	1.5	1.8	1:6	01:11	2.1	8.3
Sea Breeze	14/07/2018	90	21.4	21.6	21.4	307	327	1.5	1.7	1:2	01:19	1.9	7.6
Sea Breeze	14/07/2018	90	21.3	21.6	21.3	300	304	1.2	1.5	1:3	01:33	1.9	7.6
Sea Breeze	14/07/2018	91	21.3	21.3	21.1	291	300	0.6	1	1:1	01:41	1.2	3.3
Sea Breeze	14/07/2018	91	21	21.1	20.9	208	284	1	1.6	1	02:00	1.6	8.2
Sea Breeze	14/07/2018	91	20.9	20.9	20.8	299	295	1.4	1.6	1:5	02:07	1.9	8.1
Sea Breeze	14/07/2018	92	20.8	20.9	20.8	319	327	1.4	1.6	1:5	02:11	1.9	8.1

 blues

notehub

Discover

Help

Shelby Kaplan

Personal Access Tokens

You can have up to 10 tokens. If you reach this limit, you must delete an existing token before creating a new one or request a new token.

Token name	Created at	Expires at	Last used
Envirotech Mini Met Station	Tue 11:14:44 AM	2026-09-12 11:59:59 PM	Jul 15, 2026

Profile settings

Organization

API Access

Status

Privacy

Subscription

Sign out

Re: Request for IMS Envista API token for Sde Boker station — academic use External Inbox x

ron romov

Thank you for contacting the Israel Meteorological Service.

Token for use:

API documentation be found on the dedicated page on our website: <https://ims.gov.il/en/ObservationDataAP>

Best regards,

Ron Romo

Israel Meteorological Service

Ministry of Transport

P.O. 25 Bet Dagan 5025001 Israel

Dashboard Results

- Temperature/humidity/solar radiation track IMS's diurnal shape closely — validates the sensing approach
- The Micro Met Station lacks a proper radiation shield, and was mounted on the water building's rooftop, which absorbs and re-radiates significant heat, driving up both the SHT31 and ICP10111 readings to ~20 °C higher than Sde Boker station
- Humidity closely matched IMS despite the temperature offset — suggesting the RH sensor itself is well-calibrated and comparatively insensitive to the rooftop heat effect that's skewing temperature
- Pressure reads ~20 hPa consistently below IMS Ashalim — the fixed, stable gap (not drifting) points to an elevation/calibration offset rather than sensor noise
- Rain gauge validated by manual water-pour test during dry season
- Solar radiation follows the correct diurnal curve but reads systematically lower than Ashalim's, 15 km away under the same conditions

Envirotech Mini Met Station

429 readings · Jul 13, 01:37 PM → Jul 16, 02:07 PM · generated 2026-07-16 14:31

AIR TEMPERATURE
53.1 °C

RELATIVE HUMIDITY
29.8 %

BAROMETRIC PRESSURE
953.7 hPa

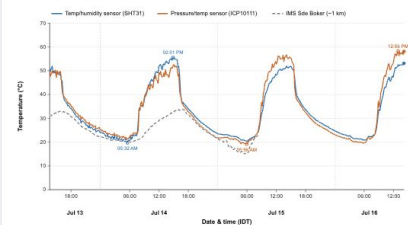
TEMP (PRESSURE SENSOR)
58.1 °C

SOLAR RADIATION
346.0 W/m²

RAINFALL (LATEST READING)
0.0 mm

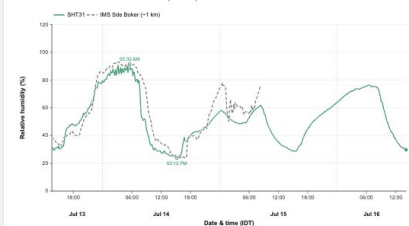
Temperature Comparison

• Temp/humidity sensor (SHT31): 53.11 °C • Pressure/temp sensor (ICP10111): 58.12 °C • IMS Sde Boker (~1 km): 22.80 °C



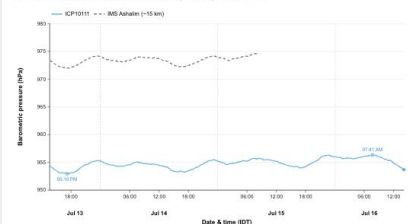
Relative Humidity

• SHT31: 29.76 % • IMS Sde Boker (~1 km): 76.00 %



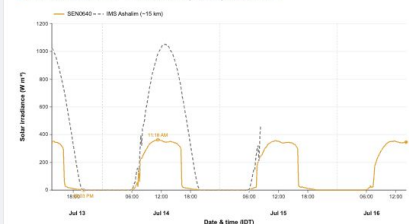
Barometric Pressure

• ICP10111: 953.69 hPa • IMS Ashalim (~15 km): 974.57 hPa



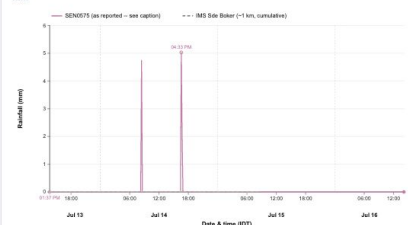
Solar Radiation

• SEN0640: 346.00 W m⁻² • IMS Ashalim (~15 km): 464.00 W m⁻²



Rainfall

• SEN0575 (as reported — see caption): 0.00 mm • IMS Sde Boker (~1 km, cumulative): 0.00 mm



Challenges & Considerations

- IMS validation:
 - Live-compared against official IMS stations — Sde Boker (~1 km, temp/humidity/rain) and Ashalim (~15 km, solar/pressure — not measured at Sde Boker) — via their public API
 - Required correcting a mislabeled timestamp offset, and
 - IMS feed has a lag and went offline temporarily for site servicing
- Rain gauge counter:
 - Resets each 10-min power cycle — reads current-cycle rain only, not a true cumulative total (needs an always-on power rail).
 - Since it's dry season, we validated it by pouring water into the gauge during an active reading window, confirming it registers real events even though it can't hold a running total

Conclusions

- Built a functioning, low-cost, solar-powered station for ~\$303
- Cellular uplink + local SD backup gives redundant, near-real-time data
- Validated against official IMS stations in a nearly-live dashboard shows realistic accuracy with known, explainable offsets
- Next steps: fix rain-gauge power wiring, install shade/radiation cover for temperature sensors, extend deployment for seasonal validation

Thank you!